

BAAZ & UDAY JKBOSE

2026 CLASS 11TH



Cell : The Unit of Life

LECTURE-4

ZOOLOGY

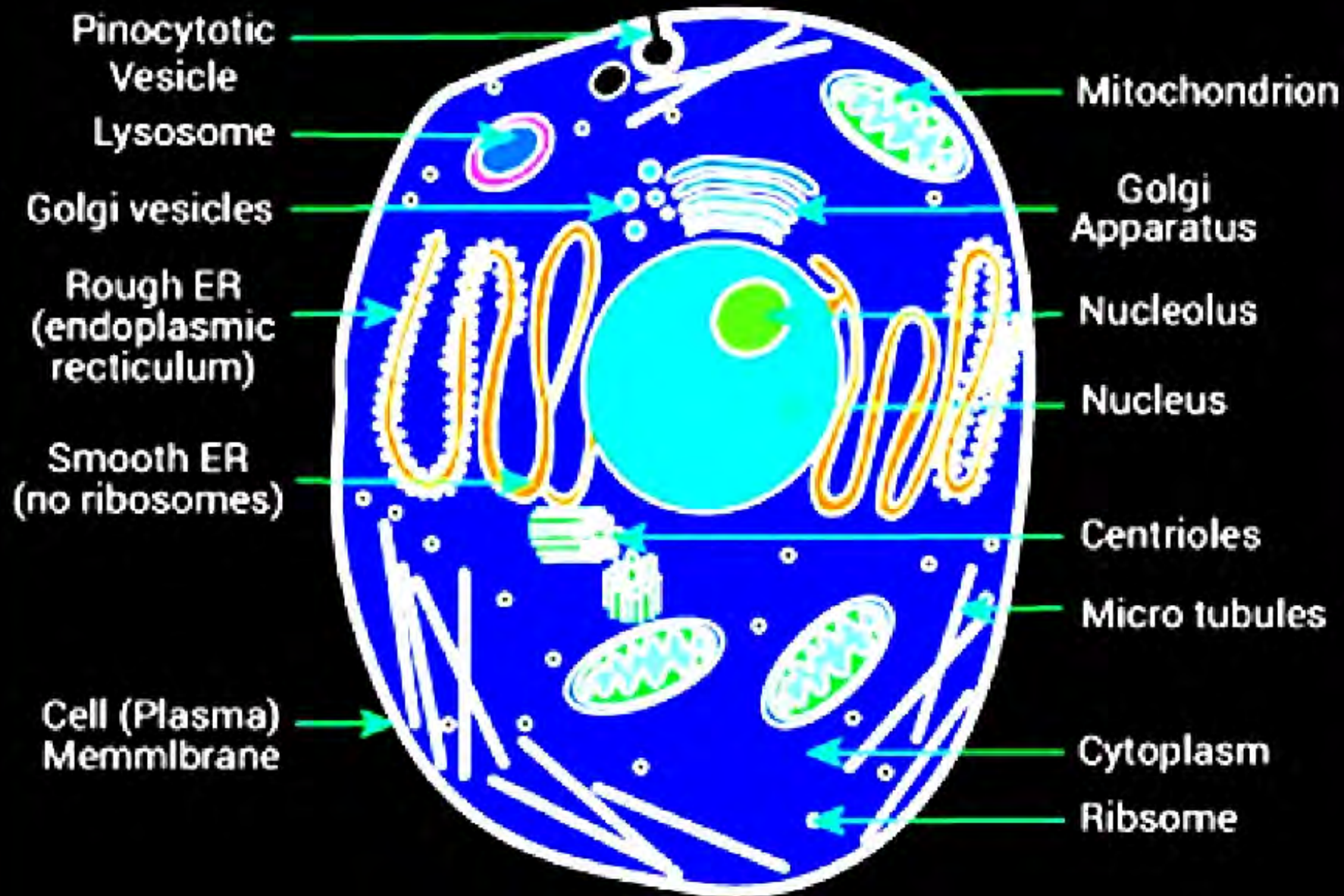
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Topics To Be

- 1 Endomembrane system (Lysosome and Vacuole)
- 2 Mitochondria
- 3 Plastid
- 4 Previous NEET Questions.

Eukaryotic Cell



Lysosomes

- These are membrane bound vesicular structures **formed by the process of packaging in the golgi apparatus.**
- The isolated lysosomal vesicles have been found to be very rich in almost all types of **hydrolytic enzymes (hydrolases like lipases, proteases, carbohydrases)** **optimally active at the acidic pH.**
- These enzymes are capable of digesting carbohydrates, proteins, lipids and nucleic acids.

Question: NEET-2019

Which of the following statements is not correct?

- 1** Lysosomes are formed by the process of packaging in the endoplasmic reticulum.
- 2** Lysosomes have numerous hydrolytic enzymes.
- 3** The hydrolytic enzymes of lysosomes are active under acidic pH.
- 4** Lysosomes are membrane-bound structures.

Vacuoles

- The vacuole is the membrane-bound space found in the cytoplasm. It contains **water, sap, excretory product and other materials not useful for the cell.**
- The vacuole is bound by a single membrane called **tonoplast**. In plant cells the vacuoles can occupy up to **90 per cent** of the volume of the cell.
- In plants, the tonoplast facilitates the transport of a number of ions and other materials against concentration gradients into the vacuole, hence their concentration is significantly higher in the vacuole than in the cytoplasm.
- In Amoeba the **contractile vacuole** is important for osmoregulation and excretion. In many cells, as in protists, **food vacuoles** are formed by engulfing the food particles.

Question: NEET-2023

Which of the following are not considered as the part of endomembrane system?

- A. Mitochondria
- B. Endoplasmic Reticulum
- C. Chloroplasts
- D. Golgi complex
- E. Peroxisomes

Choose the most appropriate answer from the options given below.

1 B and D only

2 A, D and E only

3 B and D only

4 A, C and E only

Mitochondria

- Mitochondria were first observed by **Kolliker in 1880** but **Benda in 1897** gave the present name of mitochondria.
- **Shape, number and Size:-** cylindrical or Sausage shaped and distributed in variable number depending upon the physiological activity of the cells and having a diameter of 0.5 μ m and length 2 μ m (average).
- Each mitochondrion is bound by a **double membrane, the outer membrane and the inner membrane** dividing its lumen into two aqueous compartments.
- The inner membrane is folded into ridges called **cristae**, which increase the surface area of the membrane. The inner membrane on its inner side possesses small tennis racket particles or lollipop like structure called **elementary particles** or **F₀F₁ particles** or **oxysomes**.

Mitochondria

- The space inside the inner membrane is called as inner compartment is filled with a homogeneneous substance contains called **matrix** which possesses **single circular DNA molecule, a few RNA molecules, ribosomes (70S)** and the components required for the synthesis of proteins.
- The two membranes have their own specific enzymes associated with the mitochondrial function. Mitochondria are the **sites of aerobic respiration**. They produce cellular energy in the form of **ATP**, hence they are called '**power houses**' of the cell.
- The mitochondria devide by fission.

Question: NEET-2017

Which of the following cell organelles is responsible for extracting energy from carbohydrates to form ATP?

- 1 Ribosome
- 2 Chloroplast
- 3 Mitochondrion
- 4 Lysosome



Question: NEET-2019

Which of the following statements regarding mitochondria is incorrect?

- 1 Mitochondrial matrix contains single circular DNA molecule and ribosomes.
- 2 Outer membrane is permeable to monomers of carbohydrates, fats and proteins.
- 3 Enzymes of electron transport are embedded in outer membrane.
- 4 Inner membrane is convoluted with infoldings.

Plastid



Plastid

Plastid was discovered by **Earnst Haeckel** in 1866 term Plastid was coined by Earnst Haeckel also. Plastid is a cell organelle found **only in plant cells**, Some **protists like Euglena** and are absent in animal cell.

Based on the type of pigments plastids can be classified into **chloroplasts**, **chromoplasts** and **leucoplasts**.

- The **chloroplasts** contain **chlorophyll** and **carotenoid pigments** which are responsible for trapping light energy essential for photosynthesis.
- In the chromoplasts **fat soluble carotenoid pigments** like **carotene**, **xanthophylls** and others are present. This gives the part of the plant a **yellow, orange** or **red** colour.

Plastid

The **leucoplasts** are the **colourless plastids** of varied shapes and sizes with stored nutrients and are of three types:

- **Amyloplasts:** Amyloplasts store carbohydrates (starch), e.g, potato.
- **Elaioplasts:** Elaioplasts store oils and fats.
- **Aleuroplasts:** Aleuroplasts store proteins.

Structure of a Chloroplast

Shape:- Lens shaped, Oval, spherical discoid or even ribbon like.

Size:- 5 to 10micrometer length and 2 to 4 micrometer width.

Number:- 1/cell (Chlamydomonas and green algae), 20-40/cell in mesophyll of leaves.

Like mitochondria, the chloroplasts are also double membrane bound. The inner chloroplast membrane is **less permeable**. The space limited by the inner membrane of the chloroplast is called the **stroma**. A number of organised flattened membranous sacs called the thylakoids, are present in the stroma. Thylakoids are arranged in stacks like the piles of coins called **grana**. There are also flat membranous tubules called the **stroma lamellae** connecting the thylakoids of the different grana. The membrane of the thylakoids enclose a space called a **lumen** in which chlorophyll pigments are present. The stroma of the chloroplast contains **enzymes required for the synthesis of carbohydrates and proteins**. It also contains small, **double stranded circular DNA molecules and ribosomes**. The ribosomes of the chloroplasts are smaller (70S).

Question: NEET-2024

The DNA present in chloroplast is

- 1** linear, double stranded
- 2** circular, double stranded
- 3** linear, single stranded
- 4** circular, single stranded.



Question: NEET-2024

Given below are two statements:

Statements I: Mitochondria and chloroplast both double membrane bound organelles.

Statements II: Inner membrane of mitochondria is relative less permeable, as compared chloroplast.

In the light of the above statements, choose the appropriate answer from the options given below.

- 1** Both Statement I and Statement II are correct.
- 3** Both Statement I and Statement II incorrect.
- 2** Statement I is correct but Statement II incorrect.
- 4** Statement I is incorrect but Statement II incorrect.

Question: NEET-2021

Match List – I with List – II.

List – I

- (p) **Cristae** (i) Primary constriction in chromosome
 (q) **Thylakoids** (ii) Disc-shaped sacs in Golgi apparatus
 (r) **Centromere** (iii) Infoldings in mitochondria
 (s) **Cisternae** (iv) Flattened membranous sacs in stroma of plastids

List – II

Choose the correct answer from the options given below

1 (p)-(ii), (q)-(iii), (r)-(iv), (s)-(i)

2 (p)-(iv), (q)-(iii), (r)-(ii), (s)-(i)

3 (p)-(i), (q)-(iv), (r)-(iii), (s)-(ii)

4 (p)-(iii), (q)-(iv), (r)-(i), (s)-(ii)

Question

Match List-I with List-II.

List-I

- A. Centromere** I Mitochondrion
- B. Cilium** II Cell division
- C. Cristae** III Cell movement
- D. Cisternae** IV Phospholipid Bilayer

List-II

Choose the correct answer from the options given below

1 A-II, B-I, C-IV, D-II

2 A-II, B-III, C-I, D-IV

3 A-IV, B-II, C-III, D-I

4 A-I, B-II, C-III, D-IV

Thank
You